



INFORMAL NOTES ON
MATHEMATICS
2025.10.01

微积分第五次作业

$$\begin{aligned} 4(5) \quad \lim_{x \rightarrow \frac{\pi}{4}} \tan x \tan(\frac{\pi}{4} - x) &= \frac{2 \tan x}{1 - \tan^2 x} \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x} \Big|_{x \rightarrow \frac{\pi}{4}} \\ &= \frac{2 \tan x (1 - \tan x)}{(1 - \tan x)(1 + \tan x)^2} \Big|_{x \rightarrow \frac{\pi}{4}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \tan x}{(1 + \tan x)^2} = \frac{2}{(1+1)^2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} 6 \quad 17) \quad \lim_{x \rightarrow 0} \frac{\cos x - \cos x^2}{x^2} &= \lim_{x \rightarrow 0} \frac{-2 \sin \frac{x+x^2}{2} \sin \frac{x-x^2}{2}}{x^2} = \lim_{x \rightarrow 0} \frac{-2 \frac{(x+x^2)(x-x^2)}{4}}{x^2} \\ &= \lim_{x \rightarrow 0} -\frac{(1+x)(1-x)}{2} = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 9) \quad \lim_{x \rightarrow +\infty} x(\ln(x+1) - \ln x) &= \lim_{x \rightarrow +\infty} x \ln \frac{x+1}{x} = \lim_{x \rightarrow +\infty} \ln(1 + \frac{1}{x})^x \\ &= \ln e = 1 \end{aligned}$$

$$\begin{aligned} 10 \quad 11) \quad \lim_{x \rightarrow +\infty} \frac{\ln(1+3^x)}{\ln(1+x)} &= \lim_{x \rightarrow +\infty} \frac{\ln(3^x(1+\frac{1}{3^x}))}{\ln(2^x(1+\frac{1}{2^x}))} = \lim_{x \rightarrow +\infty} \frac{x \ln 3 + \ln(1+\frac{1}{3^x})}{x \ln 2 + \ln(1+\frac{1}{2^x})} \\ &= \lim_{x \rightarrow +\infty} \frac{x \ln 3}{x \ln 2} = \frac{\ln 3}{\ln 2} \end{aligned}$$

$$\begin{aligned} 12) \quad \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt{x+1}}{\sqrt{x+1}} &= \lim_{x \rightarrow +\infty} \frac{\sqrt{1 + \frac{1}{x+1}}}{\sqrt{1 + \frac{1}{x}}} = \frac{1}{1} = 1 \end{aligned}$$

$$\begin{aligned} 15 \quad 14) \quad \lim_{x \rightarrow 0} \frac{\tan 3x}{\ln(1+x^3) - \sin 7x} &= \lim_{x \rightarrow 0} \frac{1}{\frac{\ln(1+x^3)}{\tan 3x} - \frac{\sin 7x}{\tan 3x}} = \lim_{x \rightarrow 0} \frac{1}{\frac{x^3}{3x} - \frac{7x}{3x}} \\ &= \lim_{x \rightarrow 0} \frac{1}{0 - \frac{4}{3}} = -\frac{3}{4} \end{aligned}$$

$$\begin{aligned} 20 \quad 18) \quad \cos \frac{1}{x^2} \in [-1, 1], \quad |x^2 \cos \frac{1}{x^2}| &\leq x^2, \text{ 比 } 3 \sin x \sim 3x \text{ 高阶,} \\ \therefore \lim_{x \rightarrow 0} \frac{3 \ln x + x^2 \cos \frac{1}{x^2}}{(1 + \cos x) \ln(1-3x)} &= \lim_{x \rightarrow 0} \frac{3x}{(1 + \cos x) \ln(1-3x)} = \lim_{x \rightarrow 0} \frac{3x}{2(-3x)} \\ &= \lim_{x \rightarrow 0} \frac{3x}{-6x} = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 19) \quad \lim_{x \rightarrow \infty} \left(\frac{x+1}{x+5}\right)^{x+1} &= \lim_{x \rightarrow \infty} x e^{(x+1) \ln \frac{x+1}{x+5}} = \lim_{x \rightarrow \infty} e^{(x+1) \ln(1 - \frac{4}{x+5})} \\ &= \lim_{x \rightarrow \infty} e^{x+1 \cdot (-\frac{4}{x+5})} = e^{-4} \end{aligned}$$

$$\begin{aligned} 25 \quad 23) \quad \lim_{n \rightarrow \infty} \cos^n \frac{x}{\sqrt{n}} &= \lim_{x \rightarrow \infty} e^{\frac{n \ln(\cos \frac{x}{\sqrt{n}})}{\sqrt{n}}} (1 - \frac{x^2}{2n})^n \end{aligned}$$

$$= \lim_{x \rightarrow \infty} \left(1 - \frac{x^2}{2n}\right)^{\frac{2n}{x^2}} = e^{\frac{2}{x^2}}$$

$$(25) \text{ 令 } t = \frac{1}{n(n+1)}, \quad a^{\frac{1}{n}} - a^{\frac{1}{n+1}} = a^{\frac{1}{n+1}} (a^t - 1)$$

$$\therefore \lim_{t \rightarrow 0} \frac{a^t - 1}{t} = \ln a \quad \therefore a^t - 1 = o(t/\ln a)$$

$$\therefore \lim_{n \rightarrow \infty} n^2 (\sqrt[n]{a} - \sqrt[n+1]{a}) = \lim_{n \rightarrow \infty} n^2 \cdot a^{\frac{1}{n+1}} \cdot t/\ln a$$

$$= \lim_{n \rightarrow \infty} \frac{n^2}{n(n+1)} \ln a = \ln a$$

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$$5. \text{ (1)} \quad \cos x - 1 = o\left(-\frac{x^2}{2}\right), \quad (1 + ax^2)^{\frac{1}{3}} - 1 = o\left(\frac{1}{3}ax^2\right)$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+ax^2} - 1}{\cos x - 1} = \frac{\frac{1}{3}a}{-\frac{1}{2}} = -\frac{2}{3}a = 1$$

$$\therefore a = -\frac{3}{2}$$

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8 (3) 解: $f(x) = 3x^3 - 8x^2 + x + 3$ 为多项式函数, 在 $\mathbb{R}$ 上连续.

$$f(-1) = -9, \quad f(0) = 3, \quad f(1) = -1, \quad f(2) = 15$$

$\therefore$ 在 $(-1, 0)$, $(0, 1)$, $(1, 2)$ 上各有一根.

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9. 证明: 令 $h(x) = f(x) - g(x)$, 则 h 在 $[a, b]$ 上也连续

$$h(a) = f(a) - g(a) \geq 0, \quad h(b) = f(b) - g(b) \leq 0$$

若 $h(a) = 0$, 取 $\xi = a$, 若 $h(b) = 0$, 取 $\xi = b$

若均不为 0, 由介值定理, $\exists \xi \in (a, b)$, $h(\xi) = 0$

$\therefore$ 在 $[a, b]$ 上一定存在 ξ 使 $f(\xi) = g(\xi)$

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$$11. \text{ 证明: } i \text{ 在 } (1, \lambda_2) \text{ 上时, } \lim_{x \rightarrow \lambda_1^+} \left(\frac{a_1}{x - \lambda_1} + \frac{a_2}{x - \lambda_2} + \frac{a_3}{x - \lambda_3} \right) = \lim_{x \rightarrow \lambda_1^+} \frac{a_1}{x - \lambda_1} = +\infty$$

$$\lim_{x \rightarrow \lambda_2^-} \left(\frac{a_1}{x - \lambda_1} + \frac{a_2}{x - \lambda_2} + \frac{a_3}{x - \lambda_3} \right) = \lim_{x \rightarrow \lambda_2^-} \frac{a_2}{x - \lambda_2} = -\infty$$

由介值定理, 函数在 (λ_1, λ_2) 上连续, 在 (λ_1, λ_2) 上有一根.

$$ii \text{ } (\lambda_2, \lambda_3) \text{ 上时, 同 (i) 可知 } \lim_{x \rightarrow \lambda_2^+} \left(\frac{a_1}{x - \lambda_1} + \frac{a_2}{x - \lambda_2} + \frac{a_3}{x - \lambda_3} \right) = +\infty,$$

$$\lim_{x \rightarrow \lambda_3^-} \left(\frac{a_1}{x - \lambda_1} + \frac{a_2}{x - \lambda_2} + \frac{a_3}{x - \lambda_3} \right) = -\infty, \text{ 由连续性和介值定理, 在 } (\lambda_2, \lambda_3) \text{ 上有一根.}$$

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